

pyladcp · ladcq-qa CLI guide

Process a whole cruise, step by step. Run the steps in order, in one terminal.

STEP 0 · SET THE CRUISE ROOT (FIRST!)

Every command uses `$B` = the cruise folder that holds `raw_ladcp/`, `raw_CTD/` and `sADCP/`. Set it once:

```
cd /path/to/cruise/MORIA # folder with raw_*
B="$PWD"                 # = where you are now
echo "$B"                 # MUST print that path
```

If `echo "$B"` prints nothing, STOP and set it again. An empty `$B` turns `--ladcp "$B/raw_ladcp"` into `"/raw_ladcp"` → "indexed 0 casts from 0 files". A new terminal forgets `$B` — re-set it.

YOUR FOLDER MUST LOOK LIKE THIS

```
$B/
├─ raw_ladcp/
│   ├── MASTER/ *.000 down-looker
│   └─ SLAVE/   *.000 up-looker
├─ raw_CTD/    *.hex *.hdr *.bl *.XMLCON
│               (one set per station)
└─ sADCP/      ship-ADCP (optional)
    └─ sadcp_150/DATA/ *.STA
```

Raw LADCP files carry NO station number — the CTD `.hex` (station + GPS UTC) is the anchor that tells the index which raw file is which cast.

`raw_ladcp/` may instead be a FLAT tree (no MASTER/SLAVE) — heads split by the PDO facing bit. Beam-coord PDOs are rotated to earth on read.

NEW / NON-MORIA CRUISE

`--cruise <NAME>`: any name works. An unknown cruise gets shared operator defaults (dz 8, bin-1 mask, tilt 22/4, RDI-firmware bottom track), stamped `<NAME>` on the exports.

SADCP clock not synced to GPS (wrong dates in the STA files)? Recover the offset automatically:

```
ladcp-qa ... --sadcq-timeoff auto \
--sadcq-nav "$B/Navigation"
```

slides the STA GPS against an independent nav track (SADO export or any time/lat/lon CSV). Shallow shelf cast leaking a bad near-bottom cell? raise the reject margin: `--dzbelow 24`.

STEP 1 · BUILD THE INDEX (ONCE)

Note: `--root` comes BEFORE the word `build`.

```
ladcp-index --root "$B" build \
--ladcp "$B/raw_ladcp" \
--ctd "$B/raw_CTD"
```

Expect "indexed ~40 casts from ~100 files". If it says 0 casts / 0 files → `$B` is unset (Step 0).

STEP 2 · CHECK THE PAIRING

```
ladcp-index --root "$B" show
```

Lists each station with its master + slave file. Confirm every cast got both before the long run.

STEP 3 · PROCESS THE WHOLE CRUISE

The full constrained inverse is the default solver.

```
ladcp-qa --all-stations \
--index "$B/.ladcp_archive.json" \
--root "$B" --cruise <NAME> \
--from-hex --ctd-cache "$B/ctd_from_hex" \
--sadcq "$B/sADCP/sadcq_150/DATA" \
-o "$B/qa_out"
```

A live progress bar shows X/N stations. Full detail + any failures → `$B/qa_out/ladcp-qa.log`. One bad cast is logged and skipped, not fatal.

Then open `qa_out/stations/<st>/<st>.report.pdf` → scorecard (OK / WARN / FAIL) + the figures.

STEP 4 · COMPARE VS A LEGACY REFERENCE

Have LDEO `*.mat` from a prior processing? Pair by cast time (no filename assumptions) and score:

```
ladcp-compare --ours "$B/qa_out" \
--legacy "$B/LADCP_processed" \
-o "$B/qa_out/legacy_compare"
```

→ `comparison.csv` + `comparison_report.pdf` (u/v overlays with $\pm 1\sigma$ bands). `unpaired.txt` lists any station matched on only one side — never dropped.

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Reference — what you get, every flag, and the things that trip people up.

WHAT YOU GET (UNDER \$B/QA_OUT/)

```
qa_out/
├─ stations/<MORIA-xx>/
│   ├── <st>_report.pdf      * scorecard+figs
│   ├── <st>.lad  <st>.bot velocity / b-track
│   ├── <st>.nc   <st>.xlsx NetCDF / Excel
│   ├── <st>_qa.txt/.json  QA metrics
│   └─ figures/*.png
├─ exports/                cruise-level
│   ├── MORIA_ladcp.xlsx / .nc
│   ├── MORIA_ladcp_odv.txt (Ocean Data View)
│   └─ MORIA_summary.csv
└─ ladcp-qa.log            run log
```

exports/ is written by --all-stations (or add --cruise-export to a named list of stations).

COMMON VARIATIONS

One station, full solve + ship-ADCP:

```
ladcp-qa 80 --index "$B/.ladcp_archive.json" \
--root "$B" --from-hex \
--sadc const "$B/sADCP/sadc_150/DATA" \
-o "$B/qa_out"
```

A few stations + a cruise workbook:

```
ladcp-qa 79 80 82 ... --cruise-export
```

No Excel? skip it (or install the extra):

```
ladcp-qa ... --formats odv,nc,csv
pip install pyladcp[export] # enables .xlsx
```

Stream detail instead of the bar: add -v

CONSTRAINT WEIGHTS — PHYSICAL MEANING

The inverse blends four information sources. Each weight scales how strongly that source pulls the solution relative to the LADCP water-track data:

1 = legacy-standard balance · 0 = off · >1 = trust more.

--botfac W	bottom track (default 1). Velocity over ground from the seabed echo: anchors the absolute (barotropic) level near the bottom.
--sadc const fac W	ship ADCP (default 3). The hull ADCP's absolute currents: pins the upper-ocean bins.
--barofac W	GPS navigation (default 1). Ship drift over the cast: one row pinning the depth-mean (barotropic) velocity.
--smoofac W	vertical smoothing (default 0). Curvature penalty on the profile; at 0 only ill-constrained bins are smoothed (golden).

Where each constraint acts (and how hard) is drawn per cast: figures/<st>_weights.png (legacy Fig 12).

LADCP-INDEX (--ROOT BEFORE BUILD/SHOW)

--root D	cruise root paths are relative to
build	scan + cache station→file map
--ladcp D	archive dir (has MASTER/ + SLAVE/)
--ctd D	CTD .hex/.hdr dir (the anchor)
--rescan	force full re-decode (ignore cache)
show	print the resolved casts

LADCP-QA (ORDER-FREE FLAGS)

--all-stations	every cast in the index + exports/
--cruise NAME	preset; unknown → operator defaults
--cruise-export	exports/ for a named station list
--index F	archive index → files per station
--solver S	inverse (default, full) shear
--sadc D	ship-ADCP source (inverse constraint)
--sadc const source X	vmdas (default) codas
--sadc const timeoff T	STA clock fix [s] or 'auto'
--sadc const nav P	nav track for --sadc const timeoff auto
--botfac W	bottom-track weight (see page-left)
--sadc const fac W	ship-ADCP weight (default 3)
--barofac W	GPS-barotropic weight (default 1)
--smoofac W	smoothing weight (default 0)
--dzbelow M	near-seabed reject margin [m]
--down-only	solve from the down-looker alone
--nearfield-dn-bins	device-band mask override / 'none'
--from-hex	build CTD from raw .hex if no .cnv
--ctd-cache D	reuse dir for --from-hex
--formats L	xlsx,odv,nc,csv (default: all)
-j/--jobs N	N stations in parallel (0 = auto)
-o D	output folder (default qa_out)
-v / --verbose	stream detail instead of the bar
--root D	base for cleaned CTD/ + fallback find

GOTCHAS

- Set \$B first and check echo "\$B" (Step 0).
- ladcp-index: --root goes BEFORE build / show.
- Excel needs openpyxl (pip install pyladcp[export]); else .xlsx is skipped, the rest still write.
- --from-hex needs CTD_project; a clean .cnv wins.
- Heads pair by time, so fragmented / offset master-slave deployment files still match.
- battery is an uncalibrated estimate (0.33·xmv): it can WARN but never FAILS a cast.
- nearfield_errvel_ratio WARN >1.7 = a rigid target hung below the package (e.g. a corer); mask its bins per cruise (MORIA does 03–28 automatically).
- --down-only is a cross-check — don't deliver it where the down-looker is contaminated.
- Headless server? prefix MPLBACKEND=Agg.

LADCP-COMPARE (VS LEGACY LDEO .MAT)

--ours D	ladcp-qa out dir (stations/*.nc)
--legacy D	legacy processed dir (dr *.mat)
--alt-dir D	sub a rerun in for --alt-stations
--sadc D	overlay ship-ADCP on each cast
-o D	report dir → .csv + report .pdf